E N G I N E   A R C H I T E C T U R E   R E F E R E N C E

# SENTINEL Architecture Reference

SENTINEL v2.1

Complete Signal Catalogs, Scoring Logic

All Pools, Timeframe Weights, Thresholds & Helper Functions

Confidential

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# Engine 2: SENTINEL (v2.1)

The SENTINEL engine is a trough-gated recovery scoring system that operates as a downstream confirmation engine. It can be triggered by its own multi-timeframe trough detection OR by an upstream SCOUT engine signal (score >= 75). SENTINEL uses a unique 4-timeframe architecture including a synthesized 3-Day timeframe that bridges the gap between Daily and Weekly.

## 2.1 Scoring Formula

```
core_raw = Pool_A + Pool_B + Pool_C  (capped at 100)
total_score = min(100, core_raw) + Bonus_A + Bonus_B
```

Theoretical maximum: min(100, 55+35+37) + 15 + 18 = 133. The core pool cap at 100 prevents any single pool from dominating the total score.

| Component | Max Points | Signals | Key Feature |
|---|---|---|---|
| Pool A (RSI Recovery) | 55 | 8 x 3 TFs | Flat binary reversal + live/die crossovers |
| Pool B (Price Structure) | 35 | 4 x 3 TFs | Binary: flatten, price > TP28, TP28 > WC50 |
| Pool C (Confirmation) | 37 | 7 signals | Divergence fire-and-hold + RSI > 50 gates |
| Bonus A (Structural) | 15 | 3 signals | Volume surge + support bounce + 7-component trend quality |
| Bonus B (Uptrend) | 18 | 5 signals | Gated by BB_Uptrend (MA200 rising + Close > MA200) |
| Core Cap | 100 | 19 signals | min(100, A+B+C) |
| Grand Maximum | 133 | 24 signals | 100 + 15 + 18 |

*Table 5: SENTINEL scoring structure*

## 2.2 Unique Feature: 4-Timeframe Architecture

SENTINEL is the only engine that uses 4 timeframes. The 3-Day timeframe is synthesized from Daily data by grouping every 3 daily bars into one (Open=first, High=max, Low=min, Close=last, Volume=sum). RSI, TP28, WC50, and MA200 are then recomputed from scratch on the synthesized series. This fills the gap between Daily (too noisy) and Weekly (too slow), providing smoother signals without Weekly's lag.

### 2.3 Unique Feature: Scout Gating

SENTINEL can be triggered by the SCOUT engine. When a SCOUT entry event has a score >= 75, it creates a synthetic cycle in SENTINEL that opens the scoring window. Scout-triggered cycles use raised classification thresholds: Standard Entry requires >= 80 (vs 70 for trough-triggered), and Pilot Buy requires >= 55 (vs 50). This ensures that Scout-triggered entries meet a higher bar.

### 2.4 Unique Feature: Multi-TF OR-Gate Cycle Activation

Scoring is enabled on any daily bar where ANY of these conditions is true: Daily TF is in_recovery, 4H TF is in_recovery, 3D TF is in_recovery, Weekly TF is in_recovery, or a Scout-triggered cycle is active. This means a trough on any single timeframe opens scoring across ALL timeframes, maximizing the chance of catching early recovery signals.

### 2.5 Pool A: RSI Recovery (55 pts, 8 signals x 3 TFs)

| Signal | 4H | Daily | 3D | Total | Type |
|---|---|---|---|---|---|
| A1: RSI Divergence | 1.0 | 2.0 | 2.5 | 5.5 | Fire-and-hold (multi-speed) |
| A2: RSI21 Reversal | 2.0 | 4.0 | 5.0 | 11.0 | Flat binary: RSI21 > cycle_low |
| A3: RSI36 Reversal | 1.0 | 1.5 | 2.0 | 4.5 | Flat binary: RSI36 > cycle_low |
| A4: RSI56 Reversal | 0.5 | 1.0 | 1.5 | 3.0 | Flat binary: RSI56 > cycle_low |
| A5: RSI21 > RSI36 | 2.5 | 4.0 | 5.0 | 11.5 | Live/die crossover |
| A6: RSI21 > RSI56 | 1.5 | 2.5 | 3.5 | 7.5 | Live/die crossover |
| A7: RSI36 > RSI56 | 1.0 | 2.0 | 2.5 | 5.5 | Live/die crossover |

| Signal | 4H | Daily | 3D | Total | Type |
|---|---|---|---|---|---|
| A8: All RSI > 50 | 1.5 | 3.0 | 2.0 | 6.5 | Live/die: all three above 50 |

*Table 6: SENTINEL Pool A signals with per-TF weights*

Note: SENTINEL's reversal signals (A2-A4) are flat binary, unlike SCOUT's gradual power-curve. This is a deliberate design choice: SENTINEL acts as a confirmation engine and uses simpler binary checks, while SCOUT provides the nuanced gradual scoring for early detection.

## 2.6 Pool B: Price Structure (35 pts, 4 signals x 3 TFs)

| Signal | 4H | Daily | 3D | Total | Logic |
|---|---|---|---|---|---|
| B1: TP28 Flatten | 2 | 1 | 1 | 4 | TP28 range < 10% over last 10 bars |
| B2: WC50 Flatten | 2 | 1 | 1 | 4 | WC50 range < 10% over last 10 bars |
| B3: Close > TP28 | 3 | 4 | 6 | 13 | Current close above TP28 |
| B4: TP28 > WC50 | 3 | 5 | 6 | 14 | TP28 above WC50 |

*Table 7: SENTINEL Pool B signals with per-TF weights*

## 2.7 Pool C: Confirmation (37 pts, 7 signals)

| Signal | Max | Type | Logic |
|---|---|---|---|
| C_Div_D: Daily Divergence | 8 | Fire-and-hold | Daily multi-speed divergence active |
| C_Div_4H: 4H Divergence | 5 | Fire-and-hold | 4H multi-speed divergence active |
| C_DRSI60: Daily RSI21 > 60 | 6 | Live/die | Daily RSI21 exceeds 60 |
| C_CrossTF: Cross-TF Alignment | 6 | Gated live/die | >=2 TFs with >=2 crossovers each + Price > TP28 |
| C_DRSI50: Daily RSI21 > 50 | 5 | Live/die | Daily RSI21 exceeds 50 |

| Signal | Max | Type | Logic |
|---|---|---|---|
| C_WRSI50: Weekly RSI21 > 50 | 4 | Live/die | Weekly RSI21 exceeds 50 |
| C_4HRSI50: 4H RSI21 > 50 | 3 | Live/die | 4H RSI21 exceeds 50 |

*Table 8: SENTINEL Pool C confirmation signals*

## 2.8 Bonus Pools & Classification

Bonus A (15 pts) includes Volume Surge (5), Support Bounce with two-level peak analysis (3), and a 7-component Trend Quality score (10 points max: price position, higher lows, MA200 rising, performance vs MA200, and recovery rate). Bonus B (18 pts) is gated by BB_Uptrend (MA200 rising + Close > MA200) and includes Correction Depth (5), RSI Turning Positive (3), Price Above Support (3), and Volume on Recovery (4).

SENTINEL uses score-based entry/exit: ENTRY triggers at score crossing >= 75 from below; PILOT triggers at score crossing >= 50. EXIT triggers when score drops below 40. Recovery window is 50 bars. Classification tiers: No Entry (<50), Pilot Buy (50-69), Standard Entry (70-79), Full Entry (80-89), Full Strong (90+).